YARP RV32I

Design & Verification Guide

RISC-V RV32I Processor

From ISA Fundamentals to Hardware Implementation

Key Topics:

- ✓ Computer Architecture Fundamentals
- ✓ RISC-V RV32I Instruction Set Architecture
- ✓ Instruction Formats (R, I, S, B, U, J)
- ✓ Complete ISA Reference
- ✓ Programmer's Model & Register File
- ✓ YARP Processor Microarchitecture

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1. Computer Architecture

1.1 What Is Computer Architecture?

Computer architecture describes the structure and behaviour of a computer system at a level visible to a programmer or hardware designer. It defines *how* a processor executes instructions, *how* it stores and retrieves data, and *how* the various internal components interact.

At the highest level, every computer — regardless of its complexity — follows a simple cycle:

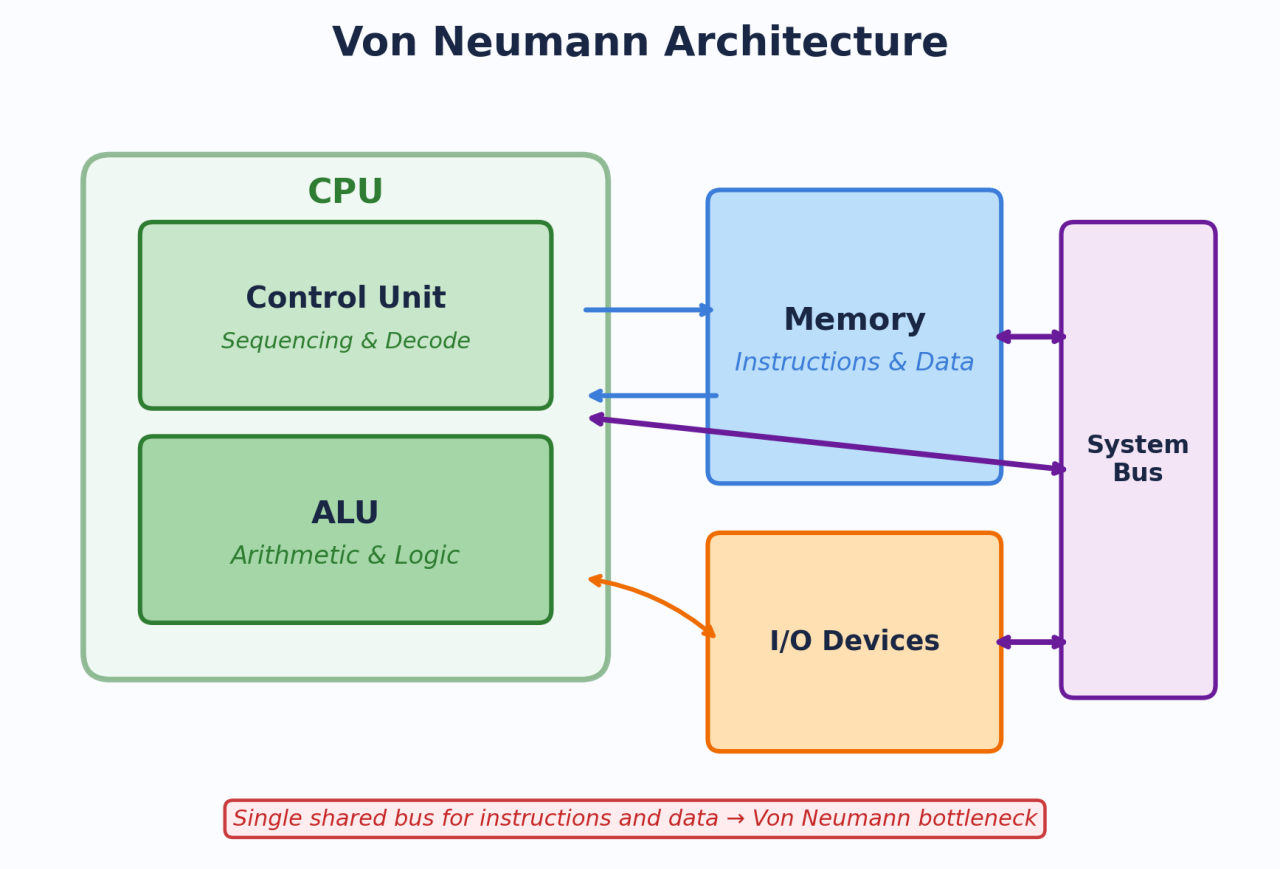
1. **Fetch** — retrieve the next instruction from memory.
2. **Decode** — interpret the instruction to determine what operation is required.
3. **Execute** — carry out the operation (arithmetic, load/store, branch, etc.).
4. **Write-back** — store the result in a register or memory.

This is commonly known as the **Fetch–Decode–Execute** cycle.

> **Key Concept:** A processor's **Instruction Set Architecture (ISA)** is the contract between software and hardware. It specifies the instructions the processor can execute, the registers available to the programmer, and the memory model. The ISA is architecture; the circuits that implement it are **microarchitecture**.

1.2 Von Neumann Architecture

The **Von Neumann architecture** (proposed by John von Neumann in 1945) is the foundational model for nearly all modern computers. Its key characteristic is a *single, shared memory* that stores both **instructions** and **data**.



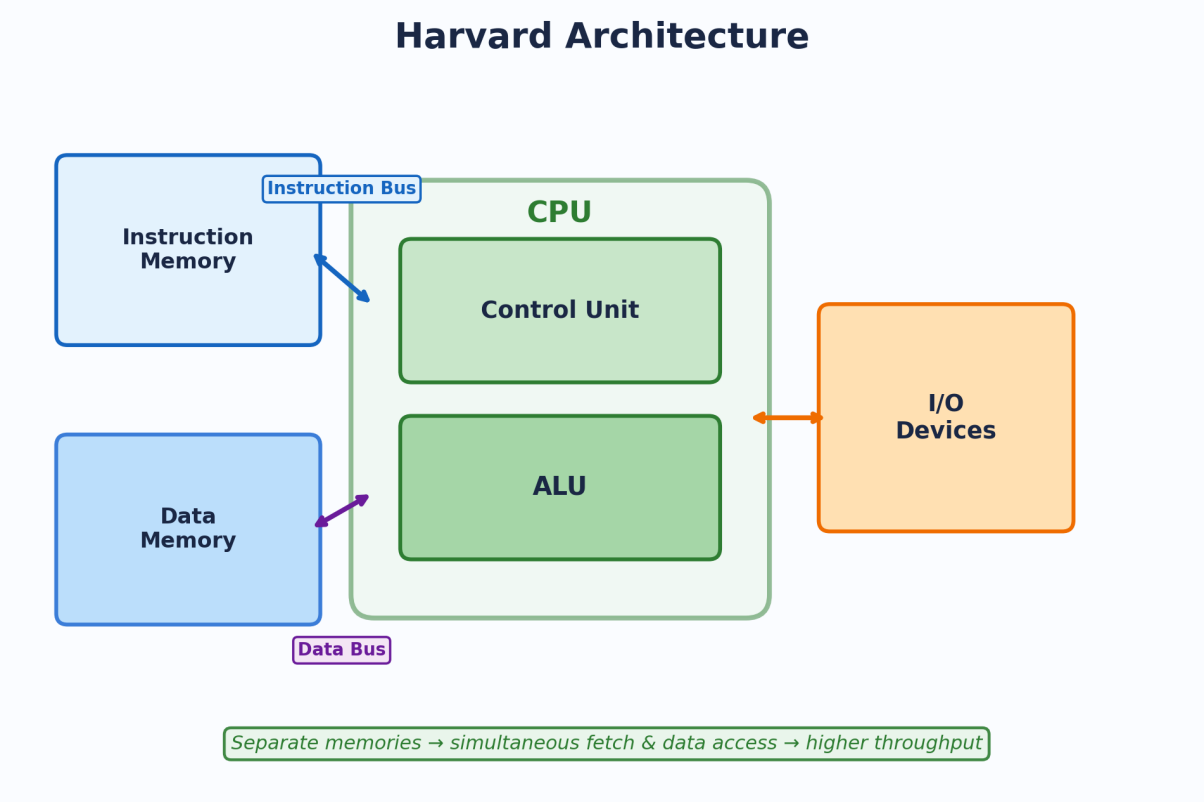
1.2.1 Key Properties

- **Single memory space:** program code and data occupy the same address range. The CPU fetches instructions and reads/writes data through the **same bus**.
- **Sequential execution:** one instruction is fetched, decoded, and executed at a time (in the simplest implementation).
- **Stored-program concept:** the program itself resides in modifiable memory, making general-purpose computing possible.

Von Neumann Bottleneck: Because instructions and data share a single bus, the CPU can either fetch an instruction *or* access data in a given cycle — never both simultaneously. This creates a bandwidth bottleneck that limits performance.

1.3 Harvard Architecture

The **Harvard architecture** solves the Von Neumann bottleneck by providing **separate memory spaces and buses** for instructions and data.



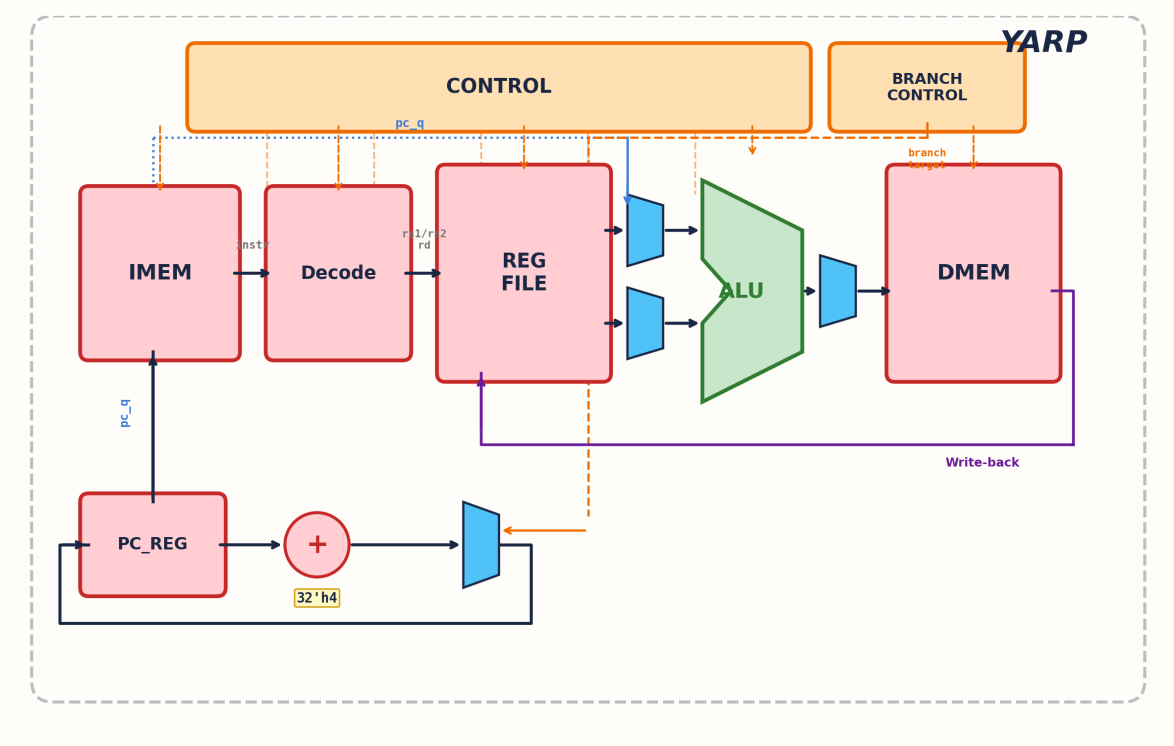
1.3.1 Key Properties

- **Separate memories:** instruction memory and data memory are physically distinct, each with its own address bus and data bus.
- **Simultaneous access:** the CPU can fetch the next instruction while reading/writing data in the same clock cycle — doubling effective bandwidth.
- **Common in embedded and RISC:** many modern processors (including RISC-V implementations) use a **Modified Harvard** architecture: separate caches for instruction and data (L1 I-cache / D-cache) backed by a unified main memory.

RISC-V and Harvard: Most practical RISC-V cores adopt a **Modified Harvard** approach — split instruction and data caches for performance, but a unified physical memory so that programs can be loaded and modified conventionally.

1.4 Inside the CPU

A CPU is built from several cooperating blocks. The diagram below shows the main components relevant to a simple RV32I implementation:



- **Program Counter (PC):** holds the address of the *current* instruction. After each fetch, the PC is incremented by 4 (the size of one RV32I instruction) or updated by a branch/jump.
- **Instruction Decoder:** breaks the 32-bit instruction word into fields (opcode, funct3, funct7, register addresses, immediate value) and determines the required operation.
- **Register File:** a bank of 32 general-purpose registers (x0–x31), each 32 bits wide. The file typically has two read ports and one write port.
- **ALU (Arithmetic Logic Unit):** performs arithmetic (add, subtract) and logic (AND, OR, XOR, shift, compare) operations.
- **Immediate Generator:** extracts and sign-extends the immediate value embedded in the instruction, rearranging the bits depending on the instruction format.
- **Control Unit:** generates control signals that steer multiplexers, the ALU operation selector, register write-enable, memory read/write, and branch decision logic.

1.5 RISC vs. CISC

Processors are broadly classified into two design philosophies:

Aspect	CISC (Complex Instruction Set)	RISC (Reduced Instruction Set)
Instruction count	Many (hundreds)	Few (typically < 50 base)
Instruction length	Variable (1–15 bytes on x86)	Fixed (4 bytes in RV32I)
Execution time	Multi-cycle per instruction	Ideally 1 cycle per instruction
Memory access	Instructions can operate directly on memory	Load/Store architecture: only `LW`/`SW` access memory
Complexity	Complex hardware decoder	Simple decoder, more compiler work
Examples	x86, x86-64	RISC-V, ARM, MIPS

Why RISC-V? RISC-V is an **open-standard** ISA released by UC Berkeley in 2010. It is modular (start with RV32I, add M, A, F, D, C extensions as needed), royalty-free, and designed for clean hardware implementation — making it an ideal educational and commercial platform.

2. The RV32I Instruction Set Architecture

2.1 Introduction to RISC-V

RISC-V (pronounced "risk-five") is a free and open Instruction Set Architecture originally developed at the University of California, Berkeley. Unlike proprietary ISAs (x86, ARM), **anyone** can implement a RISC-V processor without licensing fees.

The ISA is **modular**: a small mandatory base (`RV32I`) provides a complete integer instruction set, and optional standard extensions add functionality:

Extension	Name	Description
`I`	Base Integer	32-bit integer arithmetic, loads, stores, branches, jumps
`M`	Multiply / Divide	Hardware multiply and divide instructions
`A`	Atomic	Atomic read-modify-write for multi-core
`F`	Single-Float	IEEE 754 single-precision floating point
`D`	Double-Float	IEEE 754 double-precision floating point
`C`	Compressed	16-bit compressed instruction encoding
`Zicsr`	CSR	Control and Status Register instructions

Throughout this guide we focus exclusively on the **RV32I** base — the minimum set that every RISC-V implementation must support.

2.2 RV32I Overview

The RV32I base defines:

- **32 general-purpose registers** (`x0`–`x31`), each 32 bits wide. `x0` is hardwired to zero.
- **A 32-bit Program Counter (PC)**.
- **Fixed 32-bit instruction width** (4 bytes, naturally aligned).
- **A load/store architecture:** only load and store instructions access memory; all other operations work on registers.
- **40 base instructions** organized into 6 encoding formats.
- **Little-endian** byte ordering (with optional big-endian).
- **32-bit flat address space** ($2^{32} = 4\,\text{GiB}$).

2.3 Programmer's Model

2.3.1 The Register File

The RV32I register file contains **32 integer registers**, each 32 bits wide, designated `x0` through `x31`. In addition, each register has a *conventional ABI (Application Binary Interface) name* used by compilers and assembly programmers.

Special Register — x0 (zero): Register `x0` is **hardwired to the value 0**. Any write to `x0` is silently discarded. This simplifies many operations — for example, `addi x5, x0, 42` loads the constant 42 into `x5` without needing a separate "load immediate" instruction.

2.3.2 Register Convention (ABI)

Register	ABI Name	Role	Saver
x0	zero	Hard-wired zero	—
x1	ra	Return address	Caller
x2	sp	Stack pointer	Callee
x3	gp	Global pointer	—
x4	tp	Thread pointer	—
x5--x7	t0--t2	Temporaries	Caller
x8	s0/fp	Saved / frame pointer	Callee
x9	s1	Saved register	Callee
x10--x11	a0--a1	Function args / return vals	Caller
x12--x17	a2--a7	Function arguments	Caller
x18--x27	s2--s11	Saved registers	Callee
x28--x31	t3--t6	Temporaries	Caller

Caller-saved vs. Callee-saved: **Caller-saved** registers may be overwritten by any function call; the caller must save them if their values are needed later. **Callee-saved** registers must be preserved across function calls; if a function modifies them, it must save and restore them.

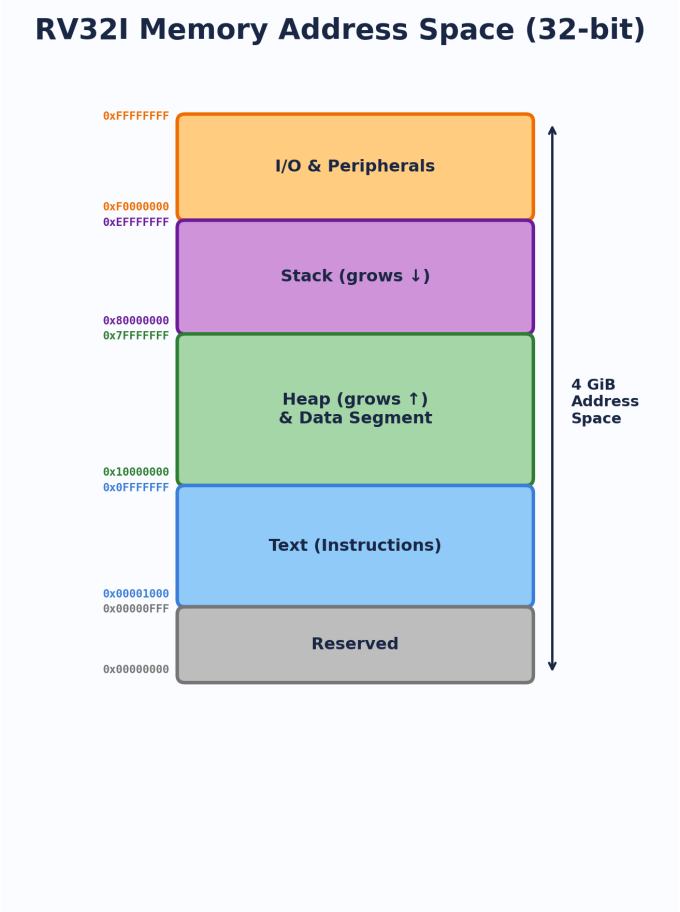
2.3.3 Program Counter

The **Program Counter (PC)** is a dedicated 32-bit register that holds the address of the *currently executing instruction*. After each instruction, the PC is updated:

- **Sequential:** $\text{PC} \leftarrow \text{PC} + 4$ (next instruction).
- **Branch taken:** $\text{PC} \leftarrow \text{PC} + \text{offset}$ (relative jump).
- **Jump:** $\text{PC} \leftarrow$ target address (absolute or register-based).

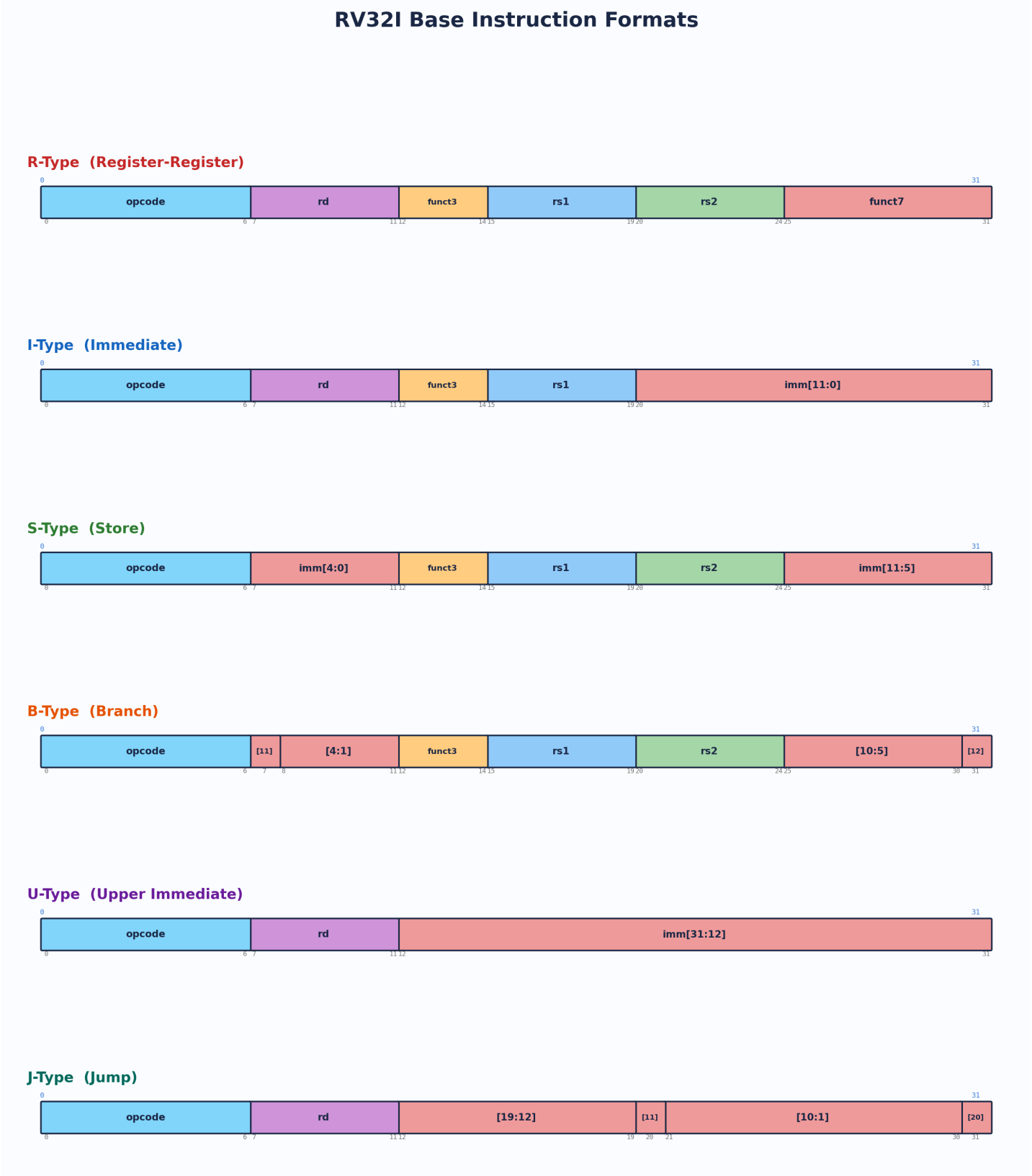
2.3.4 Memory Model

RV32I uses a **flat, byte-addressable** 32-bit address space. Memory is accessed only via load and store instructions.

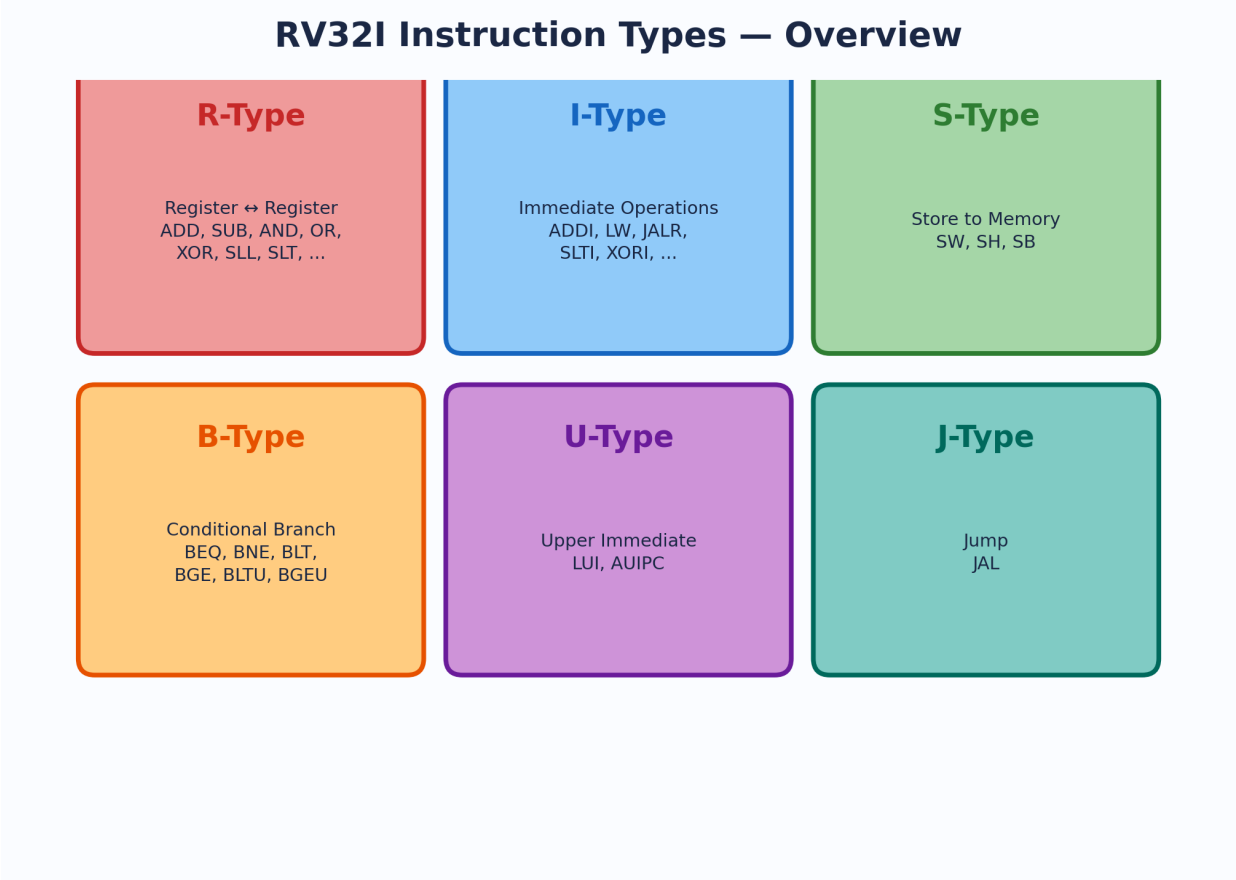

RV32I Memory Address Space (32-bit)
0xFFFFFFFF
I/O & Peripherals
0xF0000000
0xEFFFFFFF
Stack (grows ↓)
0x80000000
0x7FFFFFFF
Heap (grows ↑)
& Data Segment
0x10000000
0x0FFFFFFF
Text (Instructions)
0x00001000
0x00000FFF
Reserved
0x00000000
4 GiB
Address
Space

2.4 Instruction Formats

Every RV32I instruction is exactly **32 bits** wide. The ISA defines **six encoding formats** that specify how the 32 bits are partitioned into fields:



All formats share a common 7-bit `opcode` field in bits [6:0], which determines the general category of the instruction. The remaining 25 bits carry register addresses, function selectors, and/or immediate values.


RV32I Instruction Types — Overview
R-Type
Register ↔ Register
ADD, SUB, AND, OR,
XOR, SLL, SLT, ...
I-Type
Immediate Operations
ADDI, LW, JALR,
SLTI, XORI, ...
S-Type
Store to Memory
SW, SH, SB
B-Type
Conditional Branch
BEQ, BNE, BLT,
BGE, BLTU, BGEU
U-Type
Upper Immediate
LUI, AUIPC
J-Type
Jump
JAL

2.5 R-Type Instructions (Register–Register)

R-Type

`opcode = 0110011`

R-type instructions perform operations between **two source registers** (`rs1`, `rs2`) and write the result to a **destination register** (`rd`). No immediate value is used.

2.5.1 Format

funct7	rs2	rs1	funct3	rd	opcode
bits [31:25]	bits [24:20]	bits [19:15]	bits [14:12]	bits [11:7]	bits [6:0]
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

2.5.2 Fields

- **opcode** (7 bits): always `0110011` for R-type integer operations.
- **rd** (5 bits): destination register that receives the result.
- **funct3** (3 bits): selects the specific operation (ADD/SUB, SLL, SLT, etc.).
- **rs1** (5 bits): first source register.
- **rs2** (5 bits): second source register.
- **funct7** (7 bits): further disambiguates the operation (e.g., `0000000` for ADD vs. `0100000` for SUB).

2.5.3 R-Type Instruction Listing

Instruction	funct7	funct3	Operation
`ADD rd, rs1, rs2`	0000000	000	$\texttt{rd} \leftarrow \texttt{rs1} + \texttt{rs2}$
`SUB rd, rs1, rs2`	0100000	000	$\texttt{rd} \leftarrow \texttt{rs1} - \texttt{rs2}$
`SLL rd, rs1, rs2`	0000000	001	$\texttt{rd} \leftarrow \texttt{rs1} \ll \texttt{rs2}[4:0]$ (logical left shift)
`SLT rd, rs1, rs2`	0000000	010	$\texttt{rd} \leftarrow (\texttt{rs1} <_s \texttt{rs2})\ ?\ 1 : 0$ (signed compare)
`SLTU rd, rs1, rs2`	0000000	011	$\texttt{rd} \leftarrow (\texttt{rs1} <_u \texttt{rs2})\ ?\ 1 : 0$ (unsigned compare)
`XOR rd, rs1, rs2`	0000000	100	$\texttt{rd} \leftarrow \texttt{rs1} \oplus \texttt{rs2}$ (bitwise XOR)
`SRL rd, rs1, rs2`	0000000	101	$\texttt{rd} \leftarrow \texttt{rs1} \gg_{\text{logic}} \texttt{rs2}[4:0]$ (logical right shift)
`SRA rd, rs1, rs2`	0100000	101	$\texttt{rd} \leftarrow \texttt{rs1} \gg_{\text{arith}} \texttt{rs2}[4:0]$ (arithmetic right shift)
`OR rd, rs1, rs2`	0000000	110	$\texttt{rd} \leftarrow \texttt{rs1} \mid \texttt{rs2}$ (bitwise OR)
`AND rd, rs1, rs2`	0000000	111	$\texttt{rd} \leftarrow \texttt{rs1} \,\&\, \texttt{rs2}$ (bitwise AND)

Note: ADD and SUB share the same `funct3 = 000`. They are distinguished by `funct7`: `0000000` for ADD, `0100000` for SUB. The same pattern applies to SRL/SRA (`funct3 = 101`).

Shift reminder: A bitwise operation works directly on the 0/1 pattern stored in a register. A shift operation moves that bit pattern left or right by some number of positions.

- `SLL` **(shift left logical):** moves bits to the left, drops any bits that fall off the left side, and fills the new right-side bits with 0s. It is often used to multiply by 2^n when overflow is not a concern.
- `SRL` **(shift right logical):** moves bits to the right and fills the new left-side bits with 0s. This is useful for unsigned values and for extracting fields from an instruction or address.
- `SRA` **(shift right arithmetic):** moves bits to the right but copies the old leftmost bit (the sign bit) into the new left-side positions. This preserves the sign of a signed value.

There is no separate arithmetic left shift in RV32I: left shifts already insert 0s on the right, so `SLL` is the left-shift instruction you use.

Small example (shown in 8 bits):

$$\texttt{00101101} \ll 1 = \texttt{01011010}$$

$$\texttt{11110100} \gg_{\text{logic}} 1 = \texttt{01111010}, \qquad \texttt{11110100} \gg_{\text{arith}} 1 = \texttt{11111010}$$

If `11110100` represents -12 in two's complement, arithmetic right shift keeps the value negative, while logical right shift inserts a 0 on the left and changes the meaning.

Typical use cases: use `SLL` to scale an index by 2, 4, or 8, use `SRL` to unpack bitfields, and use `SRA` when shifting signed values right without losing the sign.

2.6 I-Type Instructions (Immediate)

I-Type

`opcodes: 0010011, 0000011, 1100111, 1110011`

I-type instructions use a **12-bit signed immediate** value as one operand alongside a source register. This format is used for **arithmetic with immediates**, **loads from memory**, **JALR**, and **system calls**.

2.6.1 Format

imm[11:0]	rs1	funct3	rd	opcode
bits [31:20]	bits [19:15]	bits [14:12]	bits [11:7]	bits [6:0]
12 bits	5 bits	3 bits	5 bits	7 bits

2.6.2 Fields

- **imm[11:0]** (12 bits): a **sign-extended** immediate value (range: −2048 to +2047).
- **rs1** (5 bits): source register.
- **funct3** (3 bits): selects the operation.
- **rd** (5 bits): destination register.
- **opcode** (7 bits): identifies the instruction sub-category.

2.6.3 I-Type Arithmetic Instructions (`opcode = 0010011`)

Instruction	funct3	Operation	
`ADDI rd, rs1, imm`	000	$\texttt{rd} \leftarrow \texttt{rs1} + \text{sext}(\text{imm})$	
`SLTI rd, rs1, imm`	010	$\texttt{rd} \leftarrow (\texttt{rs1} <_s \text{sext}(\text{imm}))\ ?\ 1 : 0$	
`SLTIU rd, rs1, imm`	011	$\texttt{rd} \leftarrow (\texttt{rs1} <_u \text{sext}(\text{imm}))\ ?\ 1 : 0$	
`XORI rd, rs1, imm`	100	$\texttt{rd} \leftarrow \texttt{rs1} \oplus \text{sext}(\text{imm})$	
`ORI rd, rs1, imm`	110	$\texttt{rd} \leftarrow \texttt{rs1} \mid \text{sext}(\text{imm})$	
`ANDI rd, rs1, imm`	111	$\texttt{rd} \leftarrow \texttt{rs1} \,\&\, \text{sext}(\text{imm})$	
`SLLI rd, rs1, shamt`	001	$\texttt{rd} \leftarrow \texttt{rs1} \ll \text{shamt}$	imm[11:5]=0000000
`SRLI rd, rs1, shamt`	101	$\texttt{rd} \leftarrow \texttt{rs1} \gg_{\text{logic}} \text{shamt}$	imm[11:5]=0000000
`SRAI rd, rs1, shamt`	101	$\texttt{rd} \leftarrow \texttt{rs1} \gg_{\text{arith}} \text{shamt}$	imm[11:5]=0100000

Shift immediates: For SLLI, SRLI, and SRAI, only the lower 5 bits of the immediate (`shamt` = shift amount, bits [4:0]) specify the shift distance (0–31). The upper bits (imm[11:5]) act like `funct7` to distinguish SRLI from SRAI.

How to think about I-Type arithmetic: Most I-type arithmetic instructions mean "take one register value and combine it with a small constant." This is why they are heavily used for loop counters, stack-pointer updates, small address adjustments, and bit masks.

- `ADDI` is the workhorse for incrementing and decrementing values.
- There is no separate `SUBI` instruction in RV32I: subtracting a constant is just `ADDI` with a negative immediate.
- `ANDI/ORI/XORI` perform bitwise operations between `rs1` and the immediate, which makes them useful for masking or setting selected bits.

Small example: `addi x5, x5, -4` subtracts 4 from `x5`.

2.6.4 I-Type Load Instructions (`opcode = 0000011`)

Load instructions read data from memory at address `rs1` + sext(imm) and place it in `rd`.

Instruction	funct3	Operation
`LB rd, imm(rs1)`	`000`	Load byte (8-bit), **sign-extend** to 32 bits → `rd`
`LH rd, imm(rs1)`	`001`	Load halfword (16-bit), **sign-extend** to 32 bits → `rd`
`LW rd, imm(rs1)`	`010`	Load word (32-bit) → `rd`
`LBU rd, imm(rs1)`	`100`	Load byte (8-bit), **zero-extend** to 32 bits → `rd`
`LHU rd, imm(rs1)`	`101`	Load halfword (16-bit), **zero-extend** to 32 bits → `rd`

Sign-extension vs. Zero-extension: `LB/LH` are **signed**: the MSB of the loaded value is replicated to fill the upper bits, preserving the sign of the original value. `LBU/LHU` are **unsigned**: the upper bits are filled with zeros. This is critical when loading narrow data into a 32-bit register.

Width reminder: In RV32I, a **byte** is 8 bits, a **halfword** is 16 bits, and a **word** is 32 bits. The load instruction tells the processor both **how many bytes to read** and **how to fill the rest of the destination register**.

- Use `LB/LBU` for 8-bit values such as characters or packed flags.
- Use `LH/LHU` for 16-bit fields.
- Use `LW` when the full 32-bit word is needed.

In a simple RV32I implementation, it is safest to think of halfwords as 2-byte aligned and words as 4-byte aligned.

2.7 S-Type Instructions (Store)

S-Type

`opcode = 0100011`

S-type instructions **store** data from a register (`rs2`) into memory. The memory address is computed as `rs1` + sext(imm). There is **no destination register** — the immediate is split across two fields.

2.7.1 Format

imm[11:5]	rs2	rs1	funct3	imm[4:0]	opcode
bits [31:25]	bits [24:20]	bits [19:15]	bits [14:12]	bits [11:7]	bits [6:0]
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

2.7.2 Why Is the Immediate Split?

The S-type immediate occupies the same bit positions as R-type's `funct7` (bits [31:25]) and `rd` (bits [11:7]). This keeps the `rs1` and `rs2` fields in exactly the same position across R, I, and S formats, simplifying the hardware register-file read logic.

2.7.3 S-Type Instruction Listing

Instruction	funct3	Operation
`SB rs2, imm(rs1)`	000	Store byte: Mem[`rs1`+sext(imm)] ← `rs2`[7 : 0]
`SH rs2, imm(rs1)`	001	Store halfword: Mem[`rs1` + sext(imm)] ← `rs2`[15 : 0]
`SW rs2, imm(rs1)`	010	Store word: Mem[`rs1` + sext(imm)] ← `rs2`[31 : 0]

Store reminder: Stores move data **from a register into memory**, so there is nothing to sign-extend or zero-extend at store time.

- `SB` writes only the lower 8 bits of `rs2`.
- `SH` writes only the lower 16 bits of `rs2`.
- `SW` writes the full 32 bits of `rs2`.

That is why RV32I has `LBU/LHU` for unsigned loads but no "unsigned store" instructions.
Small example: if `x6 = 0x12345678`, then `sb x6, 0(x5)` stores only `0x78` to memory.

2.8 B-Type Instructions (Branch)

B-Type

`opcode = 1100011`

B-type instructions implement **conditional branches**. They compare two registers (`rs1`, `rs2`) and, if the condition is true, jump to $\mathrm{PC} + \mathrm{sext}(\mathrm{offset})$. Otherwise, execution continues at $\mathrm{PC} + 4$. The offset is a **13-bit signed** value with the least significant bit always 0 (instructions are 4-byte aligned, offsets are multiples of 2).

2.8.1 Format

[12]	[10:5]	rs2	rs1	f3	[4:1]	[11]	opcode
bit 31	bits [30:25]	bits [24:20]	bits [19:15]	bits [14:12]	bits [11:8]	bit 7	bits [6:0]
1	6	5	5	3	4	1	7

2.8.2 Immediate Reconstruction

The 13-bit branch offset is assembled from the scattered fields:

$$\mathrm{offset} = \{\mathrm{imm}[12], \mathrm{imm}[11], \mathrm{imm}[10:5], \mathrm{imm}[4:1], 0\}$$

Bit [0] is always 0, giving a range of $\pm 4\,\mathrm{KiB}$ (-4096 to $+4094$).

2.8.3 B-Type Instruction Listing

Instruction	funct3	Condition (branch if true)	
`BEQ rs1, rs2, offset`	000	$\mathtt{rs1} = \mathtt{rs2}$	(equal)
`BNE rs1, rs2, offset`	001	$\mathtt{rs1} \neq \mathtt{rs2}$	(not equal)
`BLT rs1, rs2, offset`	100	$\mathtt{rs1} <_s \mathtt{rs2}$	(signed less than)
`BGE rs1, rs2, offset`	101	$\mathtt{rs1} \geq_s \mathtt{rs2}$	(signed greater or equal)
`BLTU rs1, rs2, offset`	110	$\mathtt{rs1} <_u \mathtt{rs2}$	(unsigned less than)
`BGEU rs1, rs2, offset`	111	$\mathtt{rs1} \geq_u \mathtt{rs2}$	(unsigned greater or equal)

No BLE / BGT? RISC-V omits "branch-if-less-or-equal" and "branch-if-greater-than" because they can be synthesized by swapping operands:

- `BLE rs1, rs2, offset` ≡ `BGE rs2, rs1, offset`
- `BGT rs1, rs2, offset` ≡ `BLT rs2, rs1, offset`

This keeps the hardware simple without losing expressiveness.

Branch reminder: A branch does not contain a full absolute destination address. Instead, it says "if this comparison is true, add this signed offset to the current `PC`." If the comparison is false, execution simply continues at the next instruction (`PC + 4`).

Because bit 0 of the encoded offset is always 0, the target moves in steps of 2 bytes rather than 1 byte.

Small example: if the current `PC` is `0x1000` and the branch offset is `+16`, the branch target is `0x1010`. If the condition is false, execution continues at `0x1004`.

2.9 U-Type Instructions (Upper Immediate)

U-Type

`opcodes: 0110111 (LUI), 0010111 (AUIPC)`

U-type instructions carry a **20-bit upper immediate** that is placed into bits [31:12] of the destination register. The lower 12 bits are set to zero. This format is used to construct large constants or PC-relative addresses.

2.9.1 Format

imm[31:12]	rd	opcode
bits [31:12]	bits [11:7]	bits [6:0]
20 bits	5 bits	7 bits

2.9.2 U-Type Instruction Listing

Instruction	Opcode	Operation
`LUI rd, imm`	`0110111`	`rd` $\leftarrow \text{imm}[31:12] \parallel 12'b0$ Load Upper Immediate: places the 20-bit immediate in the upper 20 bits of `rd`, zeroing the lower 12 bits.
`AUIPC rd, imm`	`0010111`	`rd` $\leftarrow \text{PC} + (\text{imm}[31:12] \parallel 12'b0)$ Add Upper Immediate to PC: adds the shifted immediate to the current PC and stores the result in `rd`. Used for PC-relative addressing of data/code.

Building 32-bit Constants: Since immediates in I-type are only 12 bits, you need **two instructions** to load a full 32-bit constant:

1. `LUI rd, upper_20` — sets bits [31:12]
2. `ADDI rd, rd, lower_12` — sets bits [11:0]

Example: load `0xDEADBEEF` into `x5`:

```
lui  x5, 0xDEADC      # x5 = 0xDEADC000
addi x5, x5, 0xEEF    # x5 = 0xDEADBEEF  (note: sign-extension adjustment)
```

LUI vs. AUIPC mental model: `LUI` builds a value from the immediate itself, while `AUIPC` builds a value **relative to the current** `PC`.

Use `LUI` when you want a large constant in a register. Use `AUIPC` when you want a base address for nearby code or data.

That is why `AUIPC` often appears with `JALR`, loads, or stores in PC-relative code sequences.

2.10 J-Type Instructions (Jump)

J-Type

`opcode = 1101111 (JAL)`

J-type provides an unconditional jump with a **21-bit signed offset** (range: ± 1 MiB). The return address ($\mathrm{PC} + 4$) is saved in `rd`, enabling function calls.

2.10.1 Format

[20]	[10:1]	[11]	[19:12]	rd	opcode
bit 31	bits [30:21]	bit 20	bits [19:12]	bits [11:7]	bits [6:0]
1	10	1	8	5	7

2.10.2 Immediate Reconstruction

The 21-bit offset is assembled as:

$$\mathrm{offset} = \{\mathrm{imm}[20], \mathrm{imm}[19:12], \mathrm{imm}[11], \mathrm{imm}[10:1], 0\}$$

2.10.3 Jump Instructions

Instruction	Type	Operation
`JAL rd, offset`	J-type	`rd` $\leftarrow \mathrm{PC} + 4$; $\mathrm{PC} \leftarrow \mathrm{PC} + \mathrm{sext}(\mathrm{offset})$ Jump And Link: saves return address in `rd`, then jumps to PC-relative target.
`JALR rd, rs1, offset`	I-type	`rd` $\leftarrow \mathrm{PC} + 4$; $\mathrm{PC} \leftarrow$ (`rs1` $+ \mathrm{sext}(\mathrm{imm})$) & ~ 1 Jump And Link Register: jumps to an address computed from a register, clearing bit 0. Used for indirect jumps and function returns.

Function call / return pattern:

- **Call:** `jal ra, function_label` — jumps to function, saves return address in `ra` (`x1`).
- **Return:** `jalr zero, ra, 0` — jumps to address in `ra`, discards link (writes to `x0`).

What does "link" mean? The **link** is just the return address, usually `PC + 4`. Saving it in `ra` lets the called function know where to come back after it finishes.

If you do not need a return address, you can discard the link by writing it to `x0`. That turns `JAL` into a plain unconditional jump.

- `jal ra, function_label` — function call
- `jal x0, label` — unconditional jump without saving a return address

`JALR` is the register-based version, which is why returns are commonly written as `jalr x0, ra, 0`.

2.11 System Instructions

System

`opcode = 1110011`

The RV32I base includes two system instructions used to request services from the execution environment (OS or debugger).

Instruction	Description
`ECALL`	Environment Call: requests a service from the operating system or execution environment (e.g., a system call). The specific service is determined by the values in `a0`–`a7`.
`EBREAK`	Environment Breakpoint: transfers control to a debugger. Used to set software breakpoints during debugging.

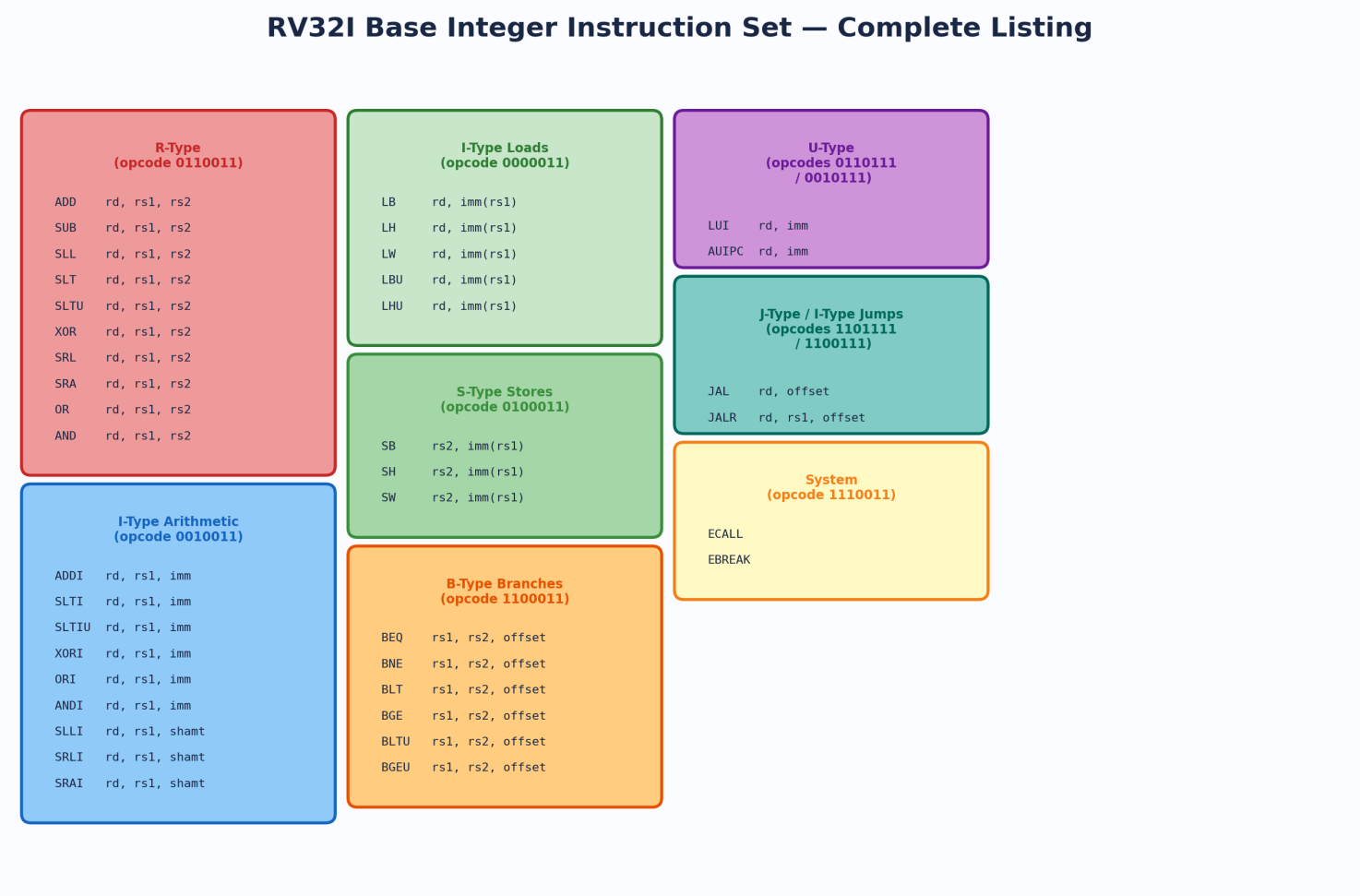
ECALL vs. EBREAK: These instructions do not behave like normal arithmetic or memory operations. Instead, they hand control to something **outside the ordinary instruction flow**.

- `ECALL` is used when the program wants the execution environment to do something on its behalf.
- `EBREAK` is used when the program wants to stop under debugger control.

The exact behavior of `ECALL` depends on the simulator, operating system, or test environment being used, while `EBREAK` is mainly a debugging aid.

3. Complete RV32I Instruction Reference

This chapter provides the **full RV32I instruction encoding table** as specified in the official RISC-V ISA manual. Every instruction is listed with its type, opcode, funct3, funct7 (where applicable), and a concise description.



3.1 Master Encoding Table

Instruction	Type	opcode	funct3	funct7/imm	Description
R-Type — Register-Register Operations (opcode 0110011)					
`ADD rd,rs1,rs2`	R	0110011	000	0000000	Add: rd = rs1 + rs2
`SUB rd,rs1,rs2`	R	0110011	000	0100000	Subtract: rd = rs1 − rs2
`SLL rd,rs1,rs2`	R	0110011	001	0000000	Shift left logical by rs2[4:0]
`SLT rd,rs1,rs2`	R	0110011	010	0000000	Set if less than (signed)
`SLTU rd,rs1,rs2`	R	0110011	011	0000000	Set if less than (unsigned)
`XOR rd,rs1,rs2`	R	0110011	100	0000000	Bitwise exclusive OR
`SRL rd,rs1,rs2`	R	0110011	101	0000000	Shift right logical by rs2[4:0]
`SRA rd,rs1,rs2`	R	0110011	101	0100000	Shift right arithmetic by rs2[4:0]
`OR rd,rs1,rs2`	R	0110011	110	0000000	Bitwise OR
`AND rd,rs1,rs2`	R	0110011	111	0000000	Bitwise AND

Continued on next page...

Instruction	Type	opcode	funct3	funct7/imm	Description
I-Type — Immediate Arithmetic (opcode 0010011)					
ADDI rd,rs1,imm	I	0010011	000	---	Add immediate
SLTI rd,rs1,imm	I	0010011	010	---	Set if $<$ imm (signed)
SLTIU rd,rs1,imm	I	0010011	011	---	Set if $<$ imm (unsigned)
XORI rd,rs1,imm	I	0010011	100	---	XOR with immediate
ORI rd,rs1,imm	I	0010011	110	---	OR with immediate
ANDI rd,rs1,imm	I	0010011	111	---	AND with immediate
SLLI rd,rs1,shamt	I	0010011	001	0000000	Shift left logical by shamt
SRLI rd,rs1,shamt	I	0010011	101	0000000	Shift right logical by shamt
SRAI rd,rs1,shamt	I	0010011	101	0100000	Shift right arithmetic by shamt
I-Type — Load Instructions (opcode 0000011)					
LB rd,imm(rs1)	I	0000011	000	---	Load byte, sign-extend
LH rd,imm(rs1)	I	0000011	001	---	Load halfword, sign-extend
LW rd,imm(rs1)	I	0000011	010	---	Load word (32-bit)
LBU rd,imm(rs1)	I	0000011	100	---	Load byte, zero-extend
LHU rd,imm(rs1)	I	0000011	101	---	Load halfword, zero-extend
S-Type — Store Instructions (opcode 0100011)					
SB rs2,imm(rs1)	S	0100011	000	---	Store byte (8-bit)
SH rs2,imm(rs1)	S	0100011	001	---	Store halfword (16-bit)
SW rs2,imm(rs1)	S	0100011	010	---	Store word (32-bit)
B-Type — Conditional Branches (opcode 1100011)					
BEQ rs1,rs2,off	B	1100011	000	---	Branch if rs1 $=$ rs2
BNE rs1,rs2,off	B	1100011	001	---	Branch if rs1 $\neq$ rs2
BLT rs1,rs2,off	B	1100011	100	---	Branch if rs1 $<$ rs2 (signed)
BGE rs1,rs2,off	B	1100011	101	---	Branch if rs1 $\geq$ rs2 (signed)
BLTU rs1,rs2,off	B	1100011	110	---	Branch if rs1 $<$ rs2 (unsigned)
BGEU rs1,rs2,off	B	1100011	111	---	Branch if rs1 $\geq$ rs2 (unsigned)
U-Type — Upper Immediate					
LUI rd,imm	U	0110111	---	---	Load upper immediate (bits [31:12])
AUIPC rd,imm	U	0010111	---	---	Add upper immediate to PC
J-Type / I-Type — Jump Instructions					
JAL rd,offset	J	1101111	---	---	Jump and link (PC-relative)
JALR rd,rs1,off	I	1100111	000	---	Jump and link register (indirect)
System Instructions (opcode 1110011)					
ECALL	I	1110011	000	000000000000	Environment call (syscall)
EBREAK	I	1110011	000	000000000001	Environment breakpoint

3.2 Instruction Count Summary

Category	Count	Instructions
R-Type (Register)	10	ADD, SUB, SLL, SLT, SLTU, XOR, SRL, SRA, OR, AND
I-Type (Imm. Arith.)	9	ADDI, SLTI, SLTIU, XORI, ORI, ANDI, SLLI, SRLI, SRAI
I-Type (Load)	5	LB, LH, LW, LBU, LHU
S-Type (Store)	3	SB, SH, SW
B-Type (Branch)	6	BEQ, BNE, BLT, BGE, BLTU, BGEU
U-Type (Upper Imm.)	2	LUI, AUIPC
J-Type (JAL)	1	JAL
I-Type (JALR)	1	JALR
System	2	ECALL, EBREAK
Total	**39**	

Design Implication: These 39 instructions (often quoted as "40 instructions" when including FENCE) represent the *entire* base RV32I ISA. From a hardware designer's perspective, each instruction translates to a specific combination of control signals. Understanding every instruction's encoding and semantics is the foundation for designing the datapath, control unit, and decoder of the YARP processor.

4. Instruction Details & Examples

This chapter provides worked examples showing how each instruction type is encoded as a 32-bit machine word and what it does conceptually.

4.1 R-Type Example: ADD x5, x6, x7

0000000	00111	00110	000	00101	0110011
funct7	x7	x6	ADD	x5	OP

Binary: 0000000 00111 00110 000 00101 0110011
Hex: 0x007302B3
Semantics: x5 ← x6 + x7. Reads registers x6 and x7, computes their sum, and writes the result to x5.

4.2 I-Type Example: ADDI x10, x0, 42

000000101010	00000	000	01010	0010011
imm = 42	x0	ADDI	x10	OP-IMM

Semantics: x10 ← x0 + 42 = 0 + 42 = 42. This is the standard way to load a small constant into a register.

4.3 S-Type Example: SW x15, 8(x2)

Store word: writes the contents of x15 to memory at address x2 + 8.

0000000	01111	00010	010	01000	0100011
imm[11:5]=0	x15	x2	SW	imm[4:0]=	STORE

Semantics: Mem[x2 + 8] ← x15

4.4 B-Type Example: BEQ x1, x2, 16

Branch to PC + 16 if x1 equals x2.
Offset = 16: binary 0 0000001000 0 → imm[12:1] = 0_000000_1000_0
The scattered immediate fields encode: imm[12]=0, imm[11]=0, imm[10:5]=000000, imm[4:1]=1000.
Semantics: If x1 = x2, then PC ← PC + 16; else PC ← PC + 4.

4.5 U-Type Example: LUI x5, 0xDEADC

Semantics: x5 ← 0xDEADC ≪ 12 = 0xDEADC000. Places the upper 20 bits directly.

4.6 J-Type Example: JAL ra, 100

Semantics:

1. `ra` $\leftarrow$ PC + 4 (save return address)
2. PC $\leftarrow$ PC + 100 (jump to target)

This is a function call: the processor remembers where to return (in `ra`) and jumps to the function body that is 100 bytes away.

4.7 Common Pseudoinstructions

RISC-V assemblers recognize **pseudoinstructions** — mnemonics that don't correspond to a single hardware instruction but are expanded into one or more real instructions:

Pseudo	Expansion	Meaning
`nop`	`addi x0, x0, 0`	No operation
`li rd, imm`	`lui + addi`	Load 32-bit immediate
`mv rd, rs`	`addi rd, rs, 0`	Copy register
`not rd, rs`	`xori rd, rs, -1`	Bitwise NOT
`neg rd, rs`	`sub rd, x0, rs`	Two's complement negate
`j offset`	`jal x0, offset`	Unconditional jump (no link)
`ret`	`jalr x0, ra, 0`	Return from function
`call label`	`auipc + jalr`	Far function call
`beqz rs, off`	`beq rs, x0, off`	Branch if zero
`bnez rs, off`	`bne rs, x0, off`	Branch if not zero
`bgt rs,rt,off`	`blt rt, rs, off`	Branch if greater than
`ble rs,rt,off`	`bge rt, rs, off`	Branch if less or equal

Why pseudoinstructions matter: Pseudoinstructions make assembly code easier to read and write, while keeping the hardware simple. The assembler transparently expands them. For verification, only the **real** (base) instructions need to be implemented in hardware.